Vector Worksheet

Grade 9 Optional Mathematics — New Syllabus 2083

Name:______________________ Roll:__________

1. Scalar and Vector Quantities

1. Define scalar quantity.
2. Define vector quantity.
3. Which of the following are scalar quantities?

   Mass, Temperature, Time, Length

4. Which of the following are vector quantities?

   Force, Velocity, Displacement, Acceleration

5. Differentiate between scalar and vector quantities.
6. Write two examples each of scalar and vector quantities.

2. Representation of Vectors

7. Represent the vector from point A to point B.
8. What is the initial point and terminal point of:

$$\vec{PQ}$$

9. Write the notation of vector joining X to Y.
10. What is meant by magnitude of a vector?
11. Draw a directed line segment representing:

$$\vec{AB}$$

3. Types of Vectors

12. Define equal vectors.
13. Define zero vector.
14. Define unit vector.
15. Define parallel vectors.
16. Define like vectors.
17. Define unlike vectors.
18. Identify:

$$\begin{bmatrix}0\\0\end{bmatrix}$$

19. Write a unit vector along x-axis.
20. Write a unit vector along y-axis.

4. Position Vector

21. Define position vector.
22. Write the position vector of point $(3, 4)$.
23. Write the position vector of point $(-2, 5)$.
24. Find the position vector of point $(0, -3)$.
25. If

$$A(2, 3)$$

write:

$$\vec{OA}$$

26. If

$$B(-4, 1)$$

write:

$$\vec{OB}$$

5. Addition of Vectors

27. Add:

$$\begin{bmatrix}2\\3\end{bmatrix} + \begin{bmatrix}1\\4\end{bmatrix}$$

28. Add:

$$\begin{bmatrix}5\\-2\end{bmatrix} + \begin{bmatrix}3\\7\end{bmatrix}$$

29. Add:

$$\begin{bmatrix}-1\\6\end{bmatrix} + \begin{bmatrix}4\\2\end{bmatrix}$$

30. Add:

$$\begin{bmatrix}7\\8\end{bmatrix} + \begin{bmatrix}-3\\5\end{bmatrix}$$

31. Verify:

$$\vec{a} + \vec{b} = \vec{b} + \vec{a}$$

for

$$\vec{a} = \begin{bmatrix}1\\2\end{bmatrix}, \quad \vec{b} = \begin{bmatrix}3\\4\end{bmatrix}$$

32. Verify:

$$(\vec{a} + \vec{b}) + \vec{c} = \vec{a} + (\vec{b} + \vec{c})$$

where

$$\vec{a} = \begin{bmatrix}1\\0\end{bmatrix}, \quad \vec{b} = \begin{bmatrix}2\\1\end{bmatrix}, \quad \vec{c} = \begin{bmatrix}3\\4\end{bmatrix}$$

6. Subtraction of Vectors

33. Subtract:

$$\begin{bmatrix}5\\7\end{bmatrix} - \begin{bmatrix}2\\3\end{bmatrix}$$

34. Subtract:

$$\begin{bmatrix}8\\4\end{bmatrix} - \begin{bmatrix}1\\6\end{bmatrix}$$

35. Subtract:

$$\begin{bmatrix}3\\9\end{bmatrix} - \begin{bmatrix}5\\2\end{bmatrix}$$

36. Does commutative property hold for subtraction? Verify.

7. Scalar Multiplication

37. Find:
$$2\begin{bmatrix}3\\4\end{bmatrix}$$

38. Find:
$$-3\begin{bmatrix}2\\5\end{bmatrix}$$

39. Find:
$$\frac{1}{2}\begin{bmatrix}8\\6\end{bmatrix}$$

40. Find:
$$4\begin{bmatrix}-1\\7\end{bmatrix}$$

41. If
$$\vec{a}=\begin{bmatrix}x\\y\end{bmatrix}$$
find:
$$5\vec{a}$$

8. Magnitude of Vector

42. Find the magnitude:
$$\begin{bmatrix}3\\4\end{bmatrix}$$

43. Find the magnitude:
$$\begin{bmatrix}5\\12\end{bmatrix}$$

44. Find the magnitude:
$$\begin{bmatrix}8\\15\end{bmatrix}$$

45. Find the magnitude:
$$\begin{bmatrix}1\\1\end{bmatrix}$$

46. Find the magnitude:
$$\begin{bmatrix}-6\\8\end{bmatrix}$$

9. Graphical Representation of Vectors

47. Represent the following vectors in arrow diagrams:

(a)
$$\vec{a}=\begin{bmatrix}2\\3\end{bmatrix}$$

(b)
$$\vec{b}=\begin{bmatrix}-3\\4\end{bmatrix}$$

(c)
$$\vec{c}=\begin{bmatrix}4\\-2\end{bmatrix}$$

(d)
$$\vec{d}=\begin{bmatrix}-5\\-4\end{bmatrix}$$

48. Find vectors $\vec{AB}$ and $\vec{BA}$:

(a)
$$A(4,3),\ B(2,5)$$

(b)
$$A(6,3),\ B(5,-4)$$

(c)
$$A(-6,3),\ B(5,-2)$$

49. If vector $\vec{AB}$ displaces point A to B, represent $\vec{AB}$ and express it as a column vector.

(a)
$$A(2,5),\ B(-1,0)$$

(b)
$$A(2,3),\ B(-5,-4)$$

(c)
$$A(-6,4),\ B(0,-1)$$

50. Draw the following position vectors:

(a)
$$\vec{OA}=\begin{bmatrix}3\\2\end{bmatrix}$$

(b)
$$\vec{OB}=\begin{bmatrix}-4\\5\end{bmatrix}$$

(c)
$$\vec{OC}=\begin{bmatrix}-2\\-3\end{bmatrix}$$

51. Write the vector joining:

(a)
$$A(1,2)\text{ to }B(4,6)$$

(b)
$$P(-3,5)\text{ to }Q(2,-1)$$

(c)
$$X(-2,-4)\text{ to }Y(3,1)$$

52. Find the magnitude:

(a)
$$\begin{bmatrix}3\\4\end{bmatrix}$$

(b)
$$\begin{bmatrix}5\\12\end{bmatrix}$$

(c)
$$\vec{AB}\quad A(1,2),\ B(4,6)$$

53. Verify graphically:
$$\vec{a}+\vec{b}=\vec{b}+\vec{a}$$
where
$$\vec{a}=\begin{bmatrix}2\\1\end{bmatrix},\quad \vec{b}=\begin{bmatrix}3\\4\end{bmatrix}$$

54. Verify triangle law using:

$$\vec{a} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}, \quad \vec{b} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

55. Draw vectors having same magnitude but opposite directions.

56. Draw two parallel vectors and two unlike vectors.

10. Mixed Problems

57. If

$$\vec{a} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \quad \vec{b} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

find:

$$\vec{a} + \vec{b}$$

58. Find:

$$2\vec{a} - \vec{b}$$

where

$$\vec{a} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}, \quad \vec{b} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

59. If

$$\vec{a} = \begin{bmatrix} x \\ 2 \end{bmatrix}, \quad \vec{b} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

and

$$\vec{a} + \vec{b} = \begin{bmatrix} 7 \\ 7 \end{bmatrix}$$

find x.

60. If

$$\vec{a} = \begin{bmatrix} 2 \\ y \end{bmatrix}, \quad 2\vec{a} = \begin{bmatrix} 4 \\ 10 \end{bmatrix}$$

find y.

61. Find the resultant vector:

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 3 \\ 4 \end{bmatrix} + \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$